Generative Project

Milestone 1 — Proposal

Flickr30k Text Embedding Baseline

Group 9 — Team and Roles

Xiaoxiao Song	Coding
Sneha Mishra	Modified the code, verified results, and drafted the proposal
Ching-Chieh Lin	Coding
Che-Wei Lee	Coding

GitHub repository: https://github.com/Sneha0504/IE7615_Generative_Project

Objective and Model

This project studies text conditioning and dataset adaptation for Stable Diffusion 1.5 (SD 1.5). The system has three pretrained components: a paired CLIP text encoder that turns a caption into a 77×768 token-embedding sequence, a latent-space U-Net that denoises the latent conditioned on those embeddings, and a VAE that decodes the final latent into a 512×512 image. Milestone 1 prepares a cleaned image–caption dataset, builds and verifies the frozen text→embedding pipeline, and runs five pretrained prompt→image sample tests. No weights are trained in this milestone.

Dataset and Preprocessing

We take a seeded (seed 42) uniform random sample across the full **31,783**-image **Flickr30k** corpus and keep **3,000** clean image–caption pairs. From the 17,500 candidate captions the notebook applies Unicode (NFKC) normalization, drops empty and duplicate captions (5 dropped), corrects EXIF orientation, de-duplicates images by pixel hash, selects one caption per image, and saves 512×512 RGB center crops. We use a custom image-disjoint **2400 / 300 / 300** train / validation / test split (no image shared across splits) rather than the Karpathy split, whose fixed 1,014 / 1,000 validation / test sets are disproportionate for a subset this size.

Executed Environment and Sample Tests

The notebook ran on **CPU** (macOS, Python 3.12.2) with PyTorch 2.6.0, Diffusers 0.35.1, and Transformers 4.49.0. All **3,000** captions were encoded to float32 77×768 sequences (tokens 6–56, median 14; **0** truncated). **18/18 validation checks passed** — split isolation, finite and repeatable embeddings, empty and long prompts, tensor reload, five saved RGB 512×512 images, and correct prompt–seed mapping. The five prompt→image tests use 512×512 pixels, DDIM with 30 steps, CFG 7.5, and seeds 42–46; the safety checker is enabled and flagged none.

ID	Prompt	Seed	Seconds	Safety
p01	A brown dog jumps over a fallen log in a green park.	42	403	clear
p02	Two people in red jackets walk along a snowy mountain trail.	43	303	clear
p03	A cyclist in a yellow shirt rides past a brick building.	44	412	clear
p04	A child in a blue coat holds a red umbrella on a rainy street.	45	493	clear
p05	Three people play basketball on an outdoor court at sunset.	46	468	clear



Five prompt→image outputs (p01–p05), in prompt order without selection.

Observations

The outputs follow the prompts closely; p04 renders a blue umbrella instead of the requested red one, a visible attribute-binding error. These five outputs are illustrative only and are not selected for quality.